

Scenario: Emergency Call Understanding System

Scenario

You are designing an AI system to assist emergency operators.

The system must:

- Classify calls (critical / urgent / non-urgent)
- Extract key information (location, symptoms, incident type)

However, you need to take into consideration the following constraints:

- There is limited and imbalanced data.
- The spoken language is Lebanese Arabic.
- The system has to work in real-time (< 2 seconds delay).
- The system needs to integrate privacy-preserving techniques.

Your Task

Design an AI system that addresses this problem while respecting the constraints.

You must justify all your choices and be ready to defend them.

Mandatory Requirements

You must integrate at least two of the following concepts in a meaningful way:

- Transfer Learning
- NLP methods
- Data Synthesis
- Privacy-preserving techniques
- Explainability

Deliverables

By the end of the session, you are expected to deliver the following:

1. Problem definition (task, inputs/outputs)
2. Pipeline (preprocessing, model, evaluation metrics)
3. External knowledge (researched concepts with justification + sources)
4. Risks and limitations